Rear Links

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System Overview

Constrain the wheels to the frame and tank the loads

Move up and down and hold up the shocks

Split into the upper (just a wishbone) and lower links (the part that holds the shocks)





The rears deal with 6 major load cases and do not have to deal with steering.

The tubes right there (might be hard to see)

System Requirements, Goals, and Constraints

Requirements:

- Adhere to geo
- Have enough clearance to the driveshaft
- Withstand the load cases

Goals:

- Increase manufacturability
  - Overall welding for the lowers
  - Make the cups more simple
- Possibly make them lighter

Past Years Design:

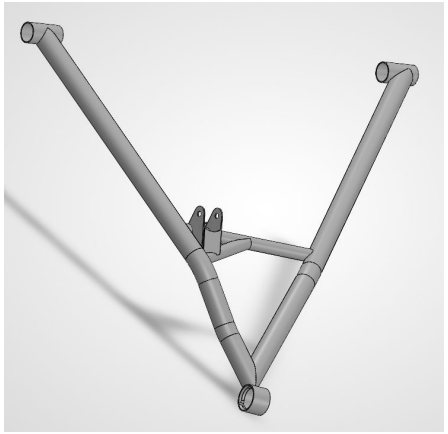
- Bent lowers
  - For more clearance to the ground
  - For better positioning on the shock tabs
- Added a crossbar
  - For a stronger braced shock tab
- Thicker trailing arm

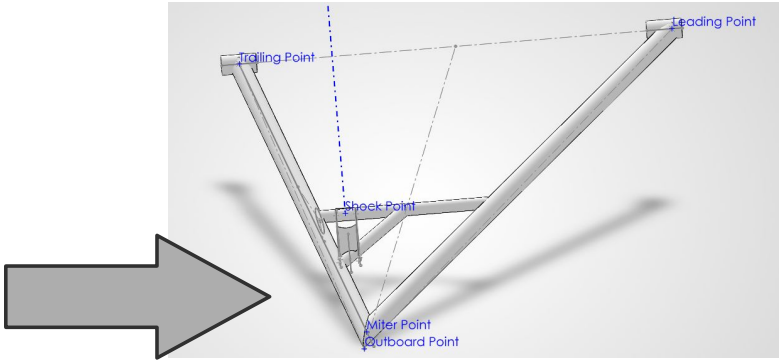
Selected Solutions

The solution I went with were the straight rears.

Why?

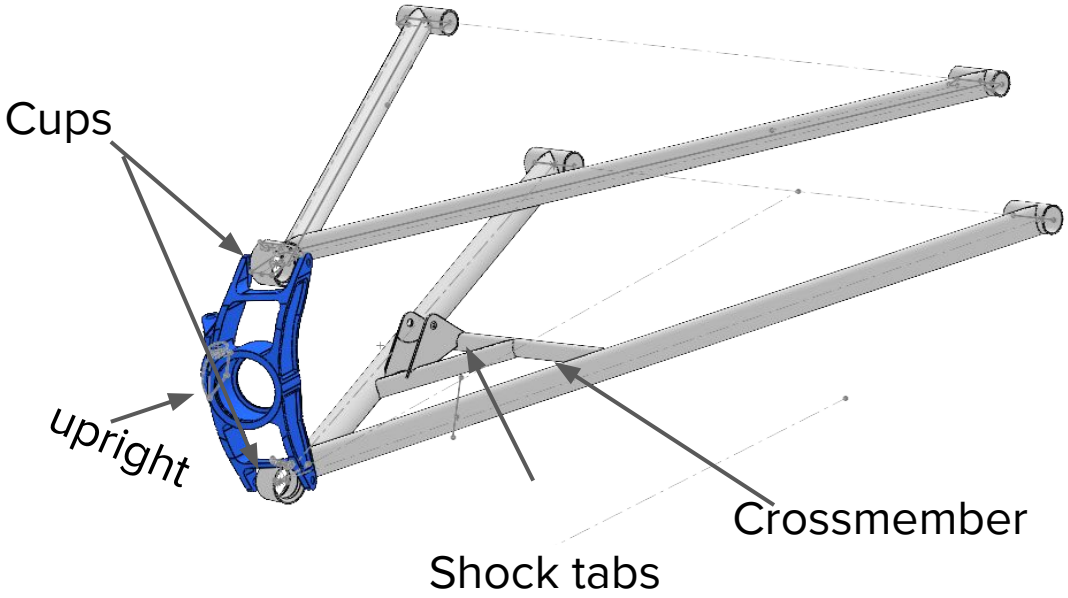
- Would not have to move the shock point on the frame (there wasn't space for a bend)
- Welding and manufacturing ease
- Remove additional stress
- Lighter



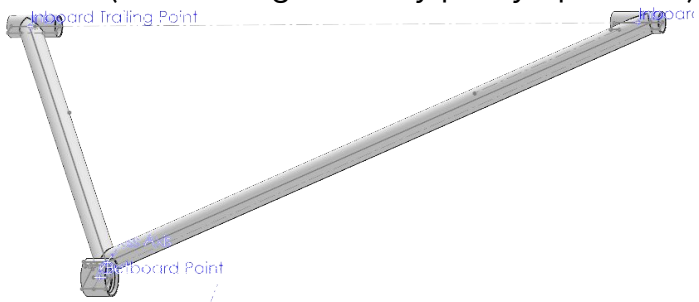


Design

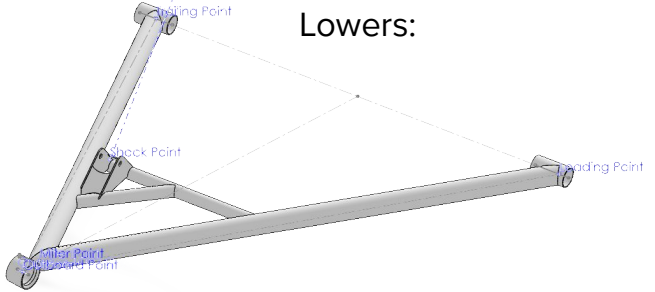
- Straight tubes for both Uppers and lowers
- Cross arm for shock tabs (increased angle for easier weld)
- The lowers are slightler thinner



Uppers:
(same design already pretty optimized)



Lowers:

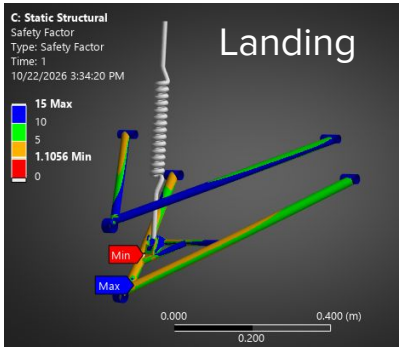


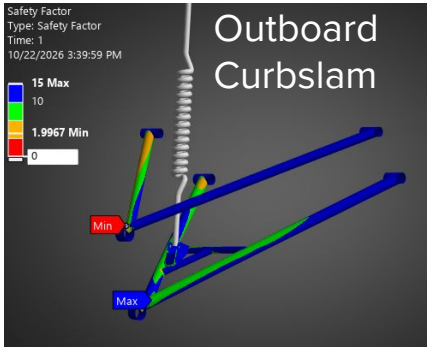
Ansys

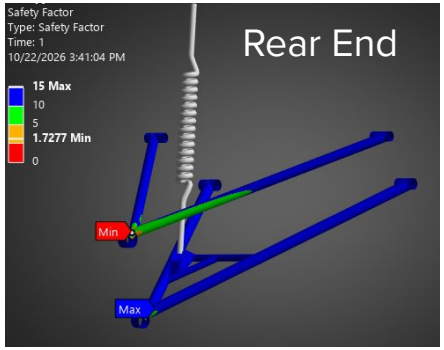
No welds for now
Welds would probably help these FOS's

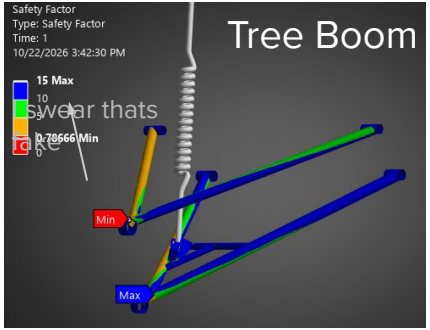
Suspension Ride position

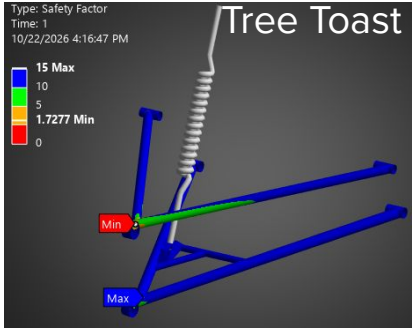
Moral of the story is work on the connection between the shock tabs and the trailing arm

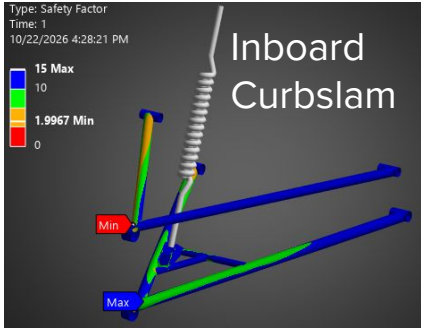












Road to FDR

- ★ Redo the ansys with welds
- ★ See what I can do about that shock tab connection
  - ○ I would appreciate ideas
- ★ Fix whatever needs to be from suggestions from CDR
- ★ Do other ansys with the other positions
- ★ Probably some testing :)

